

Q9 Implement rail fence cipher transposition operation.

```
In [1]: def rail_fence_encrypt(text, key):
    rail = [['\n' for _ in range(len(text))] for _ in range(key)]

    row, col = 0, 0
    down = True

    for ch in text:
        rail[row][col] = ch
        col += 1

        if down:
            row += 1
        else:
            row -= 1

        if row == 0:
            down = True
        if row == key - 1:
            down = False

    result = []
    for i in range(key):
        for j in range(len(text)):
            if rail[i][j] != '\n':
                result.append(rail[i][j])
    return ''.join(result)

def rail_fence_decrypt(ciphertext, key):

    rail = [['\n' for _ in range(len(ciphertext))] for _ in range(key)]

    row, col = 0, 0
    down = True
    for i in range(len(ciphertext)):
        rail[row][col] = '*'
        col += 1

        if down:
            row += 1
        else:
            row -= 1

        if row == 0:
            down = True
        if row == key - 1:
            down = False

    index = 0
    for i in range(key):
        for j in range(len(ciphertext)):
            if rail[i][j] == '*':
                rail[i][j] = ciphertext[index]
                index += 1

    result = []
    row, col = 0, 0
    down = True
    for i in range(len(ciphertext)):
        result.append(rail[row][col])
        col += 1

        if down:
            row += 1
        else:
            row -= 1

        if row == 0:
            down = True
        if row == key - 1:
            down = False

    return ''.join(result)
```

```
text = input("Enter plaintext: ")
key = int(input("Enter number of rails: "))

encrypted_text = rail_fence_encrypt(text, key)
print("Encrypted:", encrypted_text)

decrypted_text = rail_fence_decrypt(encrypted_text, key)
print("Decrypted:", decrypted_text)
```

Encrypted: HLOWRDEL OL

Decrypted: HELLO WORLD